

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Built and maintained **Java Spring Boot** backend services and REST APIs for high-availability enterprise systems, integrating with relational and document databases through JDBC and the **MongoDB Java driver**.
- Administered and performance-tuned production databases (**MongoDB, PostgreSQL, Oracle**) — designing indexes, analysing slow queries via explain plans, and cutting p95 latency on a hot collection by **38%**.
- Operated distributed database deployments with replica sets, automated backups, and monitoring; ran point-in-time recovery drills and validated failover behaviour across nodes.
- Resolved **P1/P2 production incidents in real time** using Splunk and Dynatrace, performing root-cause analysis under tight SLAs and communicating clear status updates to stakeholders during live outages.
- Acted as the technical point of contact for application teams, diagnosing their database and application issues, advising on best practices, and writing runbooks and post-incident reports that other engineers reused.
- Managed **Linux** production servers and **AWS** resources (EC2, S3, CloudWatch), configuring load balancers, firewall rules, and alerting for large-scale systems.
- Designed schema and data models for high-throughput services and ran **benchmarking** exercises to size deployments, balancing read/write patterns, indexing strategy, and storage cost.

SKILLS

Languages – Java (primary), Python, JavaScript / Node.js, SQL, Bash.

MongoDB Ecosystem – Database server, MongoDB Java & Python drivers, Atlas, Cloud Manager, Compass, automation, backup, monitoring.

Databases – MongoDB (replica sets, indexing, aggregation pipeline), PostgreSQL, Oracle, MySQL; query optimisation and performance tuning.

Distributed & Data Architecture – Replication, sharding concepts, high availability, failover, point-in-time recovery, data modelling, benchmarking.

Systems & Network Admin – Linux & Windows administration, load balancers, firewalls, networking, AWS (EC2, S3, CloudWatch), Azure fundamentals.

Diagnostics & Support – Splunk, Dynatrace, log analysis, root-cause analysis, real-time troubleshooting, SLA management, incident response.

Engineering Practices – Application architecture, Spring Boot, REST APIs, Docker, Git, CI/CD, Agile/Scrum, technical writing.

PROJECTS

Semantic Document Search on MongoDB Atlas Vector Search

- Built a **Java Spring Boot** service that generates text embeddings and stores them in **MongoDB Atlas Vector Search**, enabling semantic retrieval across a corpus of roughly 50,000 documents.
- Designed the data model and vector indexes for low-latency similarity queries, tuning candidate counts and metadata filters to keep p95 retrieval near 200ms.
- Exposed REST endpoints for ingestion and query, with health checks and CloudWatch metrics tracking throughput, latency, and error rates.

RAG Knowledge Assistant backed by MongoDB + LLM

- Developed a retrieval-augmented generation assistant in **Java** that grounds an LLM's answers in a **MongoDB-backed** knowledge base, sharply reducing hallucinated responses in internal testing.
- Implemented document chunking, embedding storage, and a re-ranking step; benchmarked retrieval quality and tuned index parameters to balance accuracy against latency.
- Containerised the service with **Docker** and deployed to **AWS**, adding structured logging and basic drift monitoring on the embedding pipeline.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore,
India**

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)